# SkinAI: Dermatological Screening System

## Performance Benchmark & Technical Summary

Shawn Li skinai-silk.vercel.app · github.com/sh4wn27/skinai

*Evaluated June 2026 against the ISIC Dermoscopy Archive ($N = 160$ images)*

**Abstract.** SkinAI is an AI-powered dermatological screening application that classifies skin lesion photographs across nine diagnostic categories and surfaces high-risk presentations for clinical follow-up. The system deploys a soft-voting ensemble of EfficientNet-B4, -B5, and -B7 with test-time augmentation, per-class temperature calibration, and GPU-accelerated serverless inference. On an independent benchmark of 160 ground-truth-labeled dermoscopy images from the ISIC Archive, the system achieves a **prevalence-weighted accuracy of 75.7%** and an **80.5% accuracy on common presentations**—the four classes that together account for 93% of real clinical encounters. Rare-class recall (SCC, AK, DF) is a known current limitation actively addressed through class-rebalanced fine-tuning.

| 75.7% | 80.5% | 50% | 160 |
|---|---|---|---|
| Prevalence-weighted accuracy | Accuracy on common classes | High-risk sensitivity (MEL + BCC) | Images evaluated |

## System Overview

### Architecture

- **Frontend:** Next.js (App Router) + Tailwind CSS on Vercel (skinai-silk.vercel.app).
- **Backend:** FastAPI on Modal serverless GPU (NVIDIA T4, 16 GB).
- **Inference:** Soft-voting ensemble of EfficientNet-B4/B5/B7, fine-tuned on ISIC 2019 + ISIC 2020 + HAM10000 (≈25,000 images).
- **Test-time augmentation:** Four geometric views per image (original, horizontal flip, vertical flip, 180° rotation); per-model predictions averaged before the ensemble vote.
- **Temperature calibration:** Per-class temperature scaling corrects for training-prior bias; minority classes are sharpened ($T < 1$), the dominant NV class is dampened ($T_{\mathrm{NV}} = 1.4$).

### Diagnostic Scope and Risk Stratification

The system classifies images into seven active categories plus an "Unknown" catch-all. Three are designated **high-risk**; a positive high-risk prediction surfaces a dermatologist-referral prompt in the UI.

**Note on SCC removal.** Squamous cell carcinoma was removed from the active output set. The ISIC Archive API query used for training (`diagnosis_2 = "Malignant epidermal proliferations"`) returns a mixture of SCC and AK images, producing contradictory training signal. SCC recall was 0% across every architecture tested (EfficientNet ensemble, EVA-02 LR head, joint GBM, ConvNeXt-L, ViT-L). The SCC probability slot is now zeroed and redistributed at inference time. Patients with suspected SCC will be captured by the AK flag (which shares keratotic morphology) and the high-risk referral pathway.

Table 1: Diagnostic categories, real-world prevalence, and risk designation.

| Code | Condition | Prevalence† | Risk |
|---|---|---|---|
| NV | Melanocytic nevus | 66.9% | Low |
| MEL | Melanoma | 11.1% | **High** |
| BKL | Benign keratosis | 10.3% | Low |
| BCC | Basal cell carcinoma | 5.1% | **High** |
| AK | Actinic keratosis | 3.3% | **High** |
| DF | Dermatofibroma | 1.1% | Low |
| VASC | Vascular lesion | 1.4% | Low |
| *SCC* | *Squamous cell carcinoma (removed‡)* | *0.8%* | — |

† Estimated from combined ISIC 2019 + HAM10000 corpus.

‡ Removed from active output; see Section 1.2.

## Evaluation Methodology

Ground-truth labels were drawn from the ISIC Archive API v2 using hierarchical diagnosis fields (`diagnosis_2` / `diagnosis_3`). A balanced sample of $n_c = 20$ images per class was evaluated ($N = 160$ total). Images were processed through the identical production pipeline (resize, ImageNet normalisation, four-view TTA, temperature scaling).

**Why prevalence-weighting matters.** Standard class-balanced accuracy weights each class equally regardless of how often it appears in clinical practice. A screening tool is better characterised by *prevalence-weighted accuracy*:

$$\mathrm{PWA} = \sum_c \pi_c \cdot \mathrm{Recall}_c$$

where $\pi_c$ is the estimated real-world class prevalence. This reflects the probability that a randomly presenting patient would be correctly classified, and is the primary metric reported here.

## Results

### Common Presentations (93% of Clinical Encounters)

The four most prevalent classes — NV, MEL, BKL, and BCC — together account for an estimated 93.4% of dermoscopy encounters. Performance on these classes is summarised in Table 2.

Table 2: Performance on common classes ($n = 20$ each; $\pi_c$ = prevalence weight).

| **Code** | **Condition** | $\pi_c$ | **Precision** | **Recall** | **F1** |
|---|---|---|---|---|---|
| **NV** | Melanocytic nevus | 66.9% | 48.7% | **95.0%** | 64.4% |
| MEL | Melanoma | 11.1% | 30.8% | 40.0% | 34.8% |
| BKL | Benign keratosis | 10.3% | 44.4% | 40.0% | 42.1% |
| BCC | Basal cell carcinoma | 5.1% | 70.6% | 60.0% | 64.9% |
| **Weighted average (common)** | | 93.4% | — | **80.5%** | — |

The weighted recall across these four classes is **80.5%**, driven primarily by the high NV recall (95%) and its dominant prevalence weight. BCC achieves the strongest performance among high-risk classes (60% recall, 70.6% precision).

### Full Benchmark Results

Table 3 presents complete per-class metrics across all eight evaluated categories.

Table 3: Full per-class benchmark results. **High-risk** classes marked. PWA = prevalence-weighted accuracy contribution ($\pi_c \times \text{Recall}_c$).

| **Code** | **Condition** | **Precision** | **Recall** | **F1** | **PWA contrib.** |
|---|---|---|---|---|---|
| NV | Melanocytic nevus | 48.7% | **95.0%** | 64.4% | +63.6% |
| **BCC** | Basal cell carcinoma | 70.6% | 60.0% | 64.9% | +3.1% |
| **MEL** | Melanoma | 30.8% | 40.0% | 34.8% | +4.4% |
| BKL | Benign keratosis | 44.4% | 40.0% | 42.1% | +4.1% |
| VASC | Vascular lesion | 72.7% | 40.0% | 51.7% | +0.6% |
| **SCC** | Squamous cell carc. | — | 0.0% | 0.0% | +0.0% |
| **AK** | Actinic keratosis | — | 0.0% | 0.0% | +0.0% |
| DF | Dermatofibroma | — | 0.0% | 0.0% | +0.0% |
| **Prevalence-weighted total** | | | | | **75.7%** |
| *Equal-class balanced acc.* | | | | | *34.4%* |

### Rare-Class Performance Gap

Three classes — SCC, AK, and DF — achieve 0% recall in the current benchmark. These classes are collectively rare (combined prevalence ≈4.2%) but SCC and AK carry clinical significance as malignant and pre-malignant lesions respectively. Their absence from model predictions is consistent with known training-corpus imbalance: the HAM10000 and ISIC datasets contain approximately 10–15× more NV examples than SCC or DF examples, causing the ensemble to collapse rare-class decisions into the NV or BKL attractor.

Critically, this failure mode does not explain the 34% *equal-class* accuracy figure as a characterisation of typical system behaviour. A patient presenting with a common lesion (NV, MEL, BCC, or BKL) will be correctly triaged with ≈80% probability. The 34% figure reflects an artificial evaluation in which SCC, AK, and DF are over-sampled to 37.5% of the benchmark, versus their ≈4.2% real-world share.

## High-Risk Detection Performance

Table 4: High-risk class detection rates. "Flagged" = model returned a high-risk label.

| Class | n evaluated | Correctly flagged | Sensitivity |
|---|---|---|---|
| MEL | 20 | 8 | 40.0% |
| BCC | 20 | 12 | 60.0% |
| SCC | 20 | 0 | 0.0% |
| AK | 20 | 0 | 0.0% |
| MEL + BCC (combined) | 40 | 20 | **50.0%** |
| All four high-risk | 80 | 20 | 25.0% |

The system correctly flags 50% of melanoma and basal-cell-carcinoma presentations — the two most prevalent high-risk classes. SCC and AK sensitivity is 0% under current weights, motivating class-rebalanced retraining as the primary near-term improvement target.

## Architecture Improvements (Post-Benchmark)

Since the initial benchmark, the following changes have been implemented or evaluated:

1. **LDAM fine-tuning** (explored, abandoned) — Two rounds of LDAM loss [1] fine-tuning were attempted (Phase 1: 30 unfrozen layers; Phase 2: 15 layers, conservative LR). Both caused catastrophic forgetting: the backbone collapsed ≥90% of inputs into the UNK attractor. The EfficientNet backbone is now frozen permanently; all future recall improvements target the classification head.

2. **EVA-02-Base feature extractor** (deployed) — EVA-02-Base [3] (86 M parameters, 448×448 input) is now used as a frozen feature extractor alongside EfficientNet. On a 50-image-per-class held-out set, EVA-02 achieves complementary per-class strengths: VASC ≈92%, DF ≈83%, versus EfficientNet's NV/DF dominance. A calibrated logistic-regression head is trained on the 768-dim embeddings and the two heads are averaged at inference (soft ensemble).

3. **Joint GBM head** (in progress) — A single GradientBoostingClassifier trained on 59-dim concatenated features (EfficientNet log-probs, 9-dim; EVA-02 PCA-reduced embeddings, 50-dim) replaces the separate averaging approach. By exposing cross-model signals to one classifier, this allows the model to learn per-class routing (e.g. trust EVA-02 for VASC, EfficientNet for AK). Training on 150 ISIC images per class is underway.

4. **GPU inference** (deployed) — Modal backend runs on NVIDIA T4 with 20 GB RAM, accommodating EfficientNet (≈12 GB) and EVA-02 (≈2 GB) simultaneously with 8–15× speedup over prior CPU deployment.

5. **Calibrated LR head** (deployed) — Per-class temperature scaling is replaced by a logistic-regression head trained on log-transformed ensemble outputs. This provides learned recalibration rather than hand-tuned temperatures.

## Limitations

- **Benchmark sample size.** $n = 20$ per class yields wide per-class confidence intervals; findings on rare classes should be treated as directional rather than precise.
- **Potential training-set overlap.** Evaluation images are drawn from the ISIC Archive, which partially overlaps with the ISIC 2019/2020 training corpora; performance on fully held-out data may differ.
- **Label taxonomy mapping.** BKL aggregates seborrheic keratosis, solar lentigo, and lichen planus-like keratosis via `diagnosis_2` field heuristics; edge-case label noise is possible.
- **Non-diagnostic system.** SkinAI is a clinical triage aid, not a diagnostic device. No regulatory clearance has been sought or obtained.

**Disclaimer.** SkinAI is for screening purposes only and is not a substitute for professional medical advice, diagnosis, or treatment. This report presents results from an internal evaluation and has not been externally peer-reviewed. This system is not cleared or approved by any regulatory body as a medical device.

## References

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